🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.4

Determine Surface Area of Prisms

Lesson 5-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the surface area of rectangular and triangular prisms.

Language Objective: I can explain my work using the words surface area, rectangular prism, triangular prism, and face.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Surface area | Rectangular prism | Triangular prism | Face | Net | Lateral face

👆 *Tap any word to see its meaning and a picture.*

1. The total area of all the faces of a solid is its ________.

2. A solid with six rectangular faces is a ________.

3. A solid with two triangular faces and three rectangular faces is a ________.

4. A flat surface of a solid figure is a ________.

5. A flat pattern that folds into a solid is a ________.

6. A side face of a prism that is not a base is a ________.

Write About the Math

How much surface area needs staining, and how many cans of stain are needed?

¿Cuánta área de superficie hay que teñir, y cuántas latas de tinte se necesitan?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Surface area** *Área de superficie*
- [ ] **Rectangular prism** *Prisma rectangular*
- [ ] **Triangular prism** *Prisma triangular*
- [ ] **Face** *Cara*
- [ ] **Net** *Plantilla (desarrollo plano)*

Model: *For both, you add the areas of all the faces, but the shapes and number of faces differ.* — Students explain both methods add up all face areas, but a rectangular prism has 6 rectangular faces while a triangular prism has 5 faces (2 triangles + 3 rectangles).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The chest's SA is ____ ft^2 because SA = 2(____×____) + 2(____×____) + 2(____×____) = ____ + ____ + ____ = ____. You need ____ can(s) because ____ ÷ 50 = ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Triangular prism
4. **Notes 4:** Face
5. **Notes 5:** Net
6. **Notes 6:** Lateral face
7. **Watch for this mistake:** *A common mistake in Surface Area of Prisms is confusing surface area with volume — multiplying $l \times w \times h$ (which finds the space INSIDE the prism, in cubic units) instead of adding the areas of all the faces ($SA = 2lw + 2lh + 2wh$, in square units). A second common slip is forgetting to double one face pair, like writing $2(l\times w) + 2(l\times h) + (w\times h)$ instead of $+ 2(w\times h)$ — always count all 6 faces (or all 5 for a triangular prism) before you submit.*
8. **Try It 1:** C. 288 in^2 — *Two triangle ends: $2 \times (½ \times 8 \times 6) = 48\ in^2$. Three rectangles: $(8 + 6 + 10) \times 10 = 24 \times 10 = 240\ in^2$. $SA = 48 + 240 = 288\ in^2$.*
9. **Try It 2:** — *A net shows every face laid out flat, so folding it up proves that surface area is just the sum of the areas of all the shapes on the net.*